# MRC Lecture 2 — Derivation Guide 4

# Multi-Head Attention: Parallel Attention Over Subspaces

Sai T. Reddy — ETH Zurich / BIIE

## 1. Motivation

In Derivation Guide 2, we derived single-head attention: one set of Q, K, V projections produces one attention pattern over the sequence. But a single attention pattern can only capture one type of relationship at a time. In molecular recognition, binding involves multiple partially independent interactions: an antibody has six CDR loops, each engaging a different region of the antigen surface. A transcription factor contacts DNA through multiple independent recognition helices. The attention mechanism needs to attend to multiple types of relationships simultaneously.

**Multi-head attention** solves this by running $H$ parallel attention computations, each with its own learned projections, operating on different subspaces of the embedding. The outputs are concatenated and linearly projected to produce the final result. This guide derives multi-head attention, connects it to the molecular correspondence with CDR loops, and shows that multi-head attention uses the same total number of parameters as a single large attention head.

## 2. Definition

**Single-head attention (recap).** With $d_k = d_{\text{model}}$:

$$\text{head} = \text{Attention}\left(X W_Q, X W_K, X W_V\right) \in \mathbb{R}^{n \times d_v}$$

where $W_Q, W_K \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and $W_V \in \mathbb{R}^{d_{\text{model}} \times d_v}$.

**Multi-head attention.** Instead of one head with $d_k = d_{\text{model}}$, use $H$ heads, each with $d_k = d_{\text{model}} / H$:

$$\text{head}_h = \text{Attention}\left(X W_Q^h, X W_K^h, X W_V^h\right) \text{for } h = 1, \ldots, H$$

where:

$$W_Q^h \in \mathbb{R}^{d_{\text{model}} \times d_k}, W_K^h \in \mathbb{R}^{d_{\text{model}} \times d_k}, W_V^h \in \mathbb{R}^{d_{\text{model}} \times d_v}$$

with $d_k = d_v = d_{\text{model}} / H$.

Concatenate the $H$ head outputs and apply a final linear projection:

$$\text{MultiHead}\left(Q, K, V\right) = \text{Concat}\left(\text{head}_1, \text{head}_2, \ldots, \text{head}_H\right) W_O$$

where $W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is the output projection matrix.

## 3. Dimension Trace

Each head operates on a subspace of dimension $d_k = d_{\text{model}} / H$.

**Example.** Let $d_{\text{model}} = 512$ and $H = 8$ heads. Then $d_k = 512 / 8 = 64$.

| Quantity | Shape per head | Shape after concatenation |
|---|---|---|
| $Q^h = X W_Q^h$ | $n \times 64$ | — |
| $K^h = X W_K^h$ | $n \times 64$ | — |
| $V^h = X W_V^h$ | $n \times 64$ | — |
| $\text{head}_h$ | $n \times 64$ | — |
| $\text{Concat}\left(\text{head}_1, \ldots, \text{head}_8\right)$ | — | $n \times 512$ |
| $\text{MultiHead} = \text{Concat} \cdot W_O$ | — | $n \times 512$ |

The concatenation of $H = 8$ heads, each of dimension 64, produces a vector of dimension $8 \times 64 = 512 = d_{\text{model}}$. The output projection $W_O$ maps this back to $d_{\text{model}}$ dimensions. The final output has the same shape ($n \times d_{\text{model}}$) as the input — this is essential for stacking multiple layers.

## 4. Parameter Count: Multi-Head = Single-Head

**Claim.** Multi-head attention with $H$ heads has the same number of parameters as single-head attention with the full $d_{\text{model}}$.

**Single-head parameters:**

- $W_Q \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$: $d_{\text{model}}^2$ parameters
- $W_K \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$: $d_{\text{model}}^2$ parameters
- $W_V \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$: $d_{\text{model}}^2$ parameters
- Total: $3\, d_{\text{model}}^2$

**Multi-head parameters:**

- Per head: $W_Q^h \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_K^h \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_V^h \in \mathbb{R}^{d_{\text{model}} \times d_k}$
- Per head: $3 \times d_{\text{model}} \times d_k$ parameters
- All heads: $H \times 3 \times d_{\text{model}} \times d_k = H \times 3 \times d_{\text{model}} \times \left(d_{\text{model}} / H\right) = 3\, d_{\text{model}}^2$
- Output projection: $W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$: $d_{\text{model}}^2$ parameters
- Total: $3\, d_{\text{model}}^2 + d_{\text{model}}^2 = 4\, d_{\text{model}}^2$

The multi-head version has one extra $d_{\text{model}}^2$ from the output projection $W_O$, but the QKV projection cost is identical. Multi-head attention does not add capacity by adding parameters — it partitions the existing capacity into independent subspaces.

**Why this matters.** The benefit of multi-head attention is not more parameters — it is more expressive attention patterns. A single head with $d_k = 512$ produces one attention distribution per query. Eight heads with $d_k = 64$ each produce eight independent attention distributions per query, each potentially attending to different positions and relationships.

## 5. Why Multiple Heads Are Needed

**The limitation of a single head.** A single attention head computes one set of attention weights per token — one probability distribution over the sequence. This means each token can express only one notion of "relevance" at a time. If token $i$ needs to simultaneously attend to a nearby token for local context AND a distant token for long-range dependency, a single head must compromise.

**Multiple heads resolve this.** Each head learns its own projection matrices, which means each head can attend to different aspects of the input:

- **Head 1** might specialize in local context (attending to nearby positions)
- **Head 2** might specialize in long-range dependencies (attending to distant positions)
- **Head 3** might specialize in syntactic relationships (subject-verb agreement in text, or secondary structure contacts in proteins)
- **Head 4** might specialize in semantic relationships (coreference in text, or functional motifs in proteins)

The concatenation of all heads provides a rich, multi-faceted representation that captures multiple types of relationships simultaneously.

**Empirical evidence.** Visualization of attention patterns in trained transformers (Vaswani et al., 2017; Clark et al., 2019) confirms that different heads learn qualitatively different attention patterns. Some heads attend locally, others attend to specific syntactic positions, and others show broad, diffuse attention. For protein language models (Rives et al., 2021), attention heads have been shown to capture residue-residue contacts — different heads capturing contacts at different spatial ranges.

## 6. The Molecular Correspondence: CDR Loops as Attention Heads

In the Computational Convergence framework (Reddy, bioRxiv, 2026), multi-head attention has a direct physical correspondence with the structure of molecular recognition interfaces.

**Antibody-antigen binding.** An antibody recognizes its target antigen through six CDR (Complementarity-Determining Region) loops: CDRH1, CDRH2, CDRH3 on the heavy chain, and CDRL1, CDRL2, CDRL3 on the light chain. Each CDR loop engages a different region of the antigen epitope surface. The loops are structurally distinct (different lengths, conformations, amino acid compositions) and make partially independent contributions to binding.

**The correspondence:**

| Antibody structure | Multi-head attention | Role |
|---|---|---|
| 6 CDR loops | $H$ attention heads | Parallel recognition units |
| Each loop has distinct sequence and structure | Each head has distinct $W_Q^h, W_K^h, W_V^h$ | Different learned projections |
| Different loops contact different epitope regions | Different heads attend to different positions | Specialized attention patterns |
| Loops contribute different interaction types (charge, hydrophobic, H-bond) | Heads specialize for different relationships | Diverse feature extraction |
| Combined paratope = union of all CDR contributions | Combined output = Concat($\text{head}_1$, …, head_H) | Multi-faceted representation |
| Total binding = sum of CDR energies (approx.) | Total representation = concatenation of head outputs | Aggregation |

**Important caveat.** This correspondence is structural, not exact. There is no one-to-one mapping between a specific CDR loop and a specific attention head. The number of heads ($H$) need not equal 6. The correspondence is that both systems use multiple partially independent recognition channels that are combined into a single binding or representation outcome. The attention mechanism's ability to learn multiple independent attention patterns from data mirrors the immune system's use of multiple CDR loops with distinct binding contributions.

**Extension to other domains.** The multi-head correspondence generalizes:

- **TCR-pMHC:** TCR has 6 CDR loops (analogous to antibody), contacting both the peptide and the MHC groove
- **TF-DNA:** Some TFs use multiple zinc fingers or recognition helices, each contacting different DNA positions
- **Enzyme-substrate:** Active sites often have multiple subsites (S1, S2, S1', S2' in protease nomenclature), each engaging different substrate positions

In each case, the physical binding interface has multiple partially independent contact regions, and multi-head attention provides the computational machinery to represent this multiplicity.

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## 7. Worked Example: Two Heads vs. One Head

**Problem.** Compare single-head and two-head attention for $n=3$ tokens with $d_{\text{model}}=4$.

**Single head ($H=1$, $d_k=4$)**

Suppose the score matrix (after scaling) is:

$$\tilde{S}=\begin{pmatrix}1.0 & 0.5 & 0.8\\ 0.3 & 1.0 & 0.2\\ 0.9 & 0.4 & 1.0\end{pmatrix}$$

After row-wise softmax:

$$A=\begin{pmatrix}0.401 & 0.243 & 0.328\\ 0.271 & 0.545 & 0.245\\ 0.377 & 0.229 & 0.417\end{pmatrix}$$

Each token has one attention distribution over the sequence.

**Two heads ($H=2$, $d_k=2$ each)**

Head 1 projects into a 2D subspace capturing one type of relationship. Suppose its score matrix (after scaling) is:

$$\tilde{S}^{(1)}=\begin{pmatrix}0.5 & 2.0 & 0.1\\ 0.3 & 0.3 & 1.5\\ 1.8 & 0.2 & 0.4\end{pmatrix}$$

After softmax:

$$A^{(1)}=\begin{pmatrix}0.163 & 0.730 & 0.109\\ 0.201 & 0.201 & 0.667\\ 0.637 & 0.129 & 0.157\end{pmatrix}$$

Head 2 projects into a different 2D subspace. Suppose its score matrix is:

$$\tilde{S}^{(2)}=\begin{pmatrix}1.5 & 0.1 & 0.2\\ 0.8 & 0.8 & 0.1\\ 0.3 & 0.9 & 1.2\end{pmatrix}$$

After softmax:

$$A^{(2)}=\begin{pmatrix}0.576 & 0.142 & 0.157\\ 0.372 & 0.372 & 0.185\\ 0.183 & 0.333 & 0.449\end{pmatrix}$$

**Comparison of attention patterns:**

| Token | Single head | Head 1 | Head 2 |
|---|---|---|---|
| Token 1 attends to… | Token 1 (0.40) | **Token 2 (0.73)** | **Token 1 (0.58)** |
| Token 2 attends to… | Token 2 (0.55) | **Token 3 (0.67)** | **Tokens 1,2 (0.37 each)** |
| Token 3 attends to… | Token 3 (0.42) | **Token 1 (0.64)** | **Token 3 (0.45)** |

The single head produces a blurred compromise pattern. The two heads produce distinct, specialized patterns: Head 1 attends primarily to different tokens than Head 2. After concatenation and projection by $W_O$, the network combines both types of information.

In molecular terms: Head 1 might capture short-range contacts (like CDRH3 engaging the epitope center) while Head 2 captures long-range interactions (like CDRL1 engaging the epitope periphery). The single-head version would blur these into one averaged pattern.

## 8. Summary

| Concept | Definition | Significance |
|---|---|---|
| Multi-head attention | $H$ parallel attention heads, concatenated + projected | Multiple independent attention patterns |
| Per-head dimension | $d_k=d_{\text{model}}/H$ | Subspace partitioning |
| Output | $\text{Concat}\left(\text{head}_1,\dots,\text{head}_H\right)W_O$ | Combined multi-faceted representation |
| Parameter cost | Same QKV parameters as single head | Expressiveness without extra cost |
| Head specialization | Different heads learn different patterns | Local vs. long-range, syntactic vs. semantic |
| CDR correspondence | 6 CDR loops ↔ $H$ attention heads | Partially independent recognition channels |

## 9. Checkpoint Questions

**Q1.** A transformer has $d_{\text{model}} = 768$ and $H = 12$ heads. What is $d_k$ per head? How many total parameters are in the QKV projections (excluding $W_O$)?
*Answer:* $d_k = 768 / 12 = 64$. QKV parameters: $3 \times d_{\text{model}}^2 = 3 \times 768^2 = 1,769,472$. This is the same as a single head with $d_k = 768$.
**Q2.** If you increase the number of heads from $H = 8$ to $H = 16$ while keeping $d_{\text{model}} = 512$, what happens to $d_k$ per head? What is the tradeoff?
*Answer:* $d_k$ drops from 64 to 32. Each head operates in a smaller subspace, which limits the complexity of relationships it can capture within that subspace. But there are now 16 independent attention patterns instead of 8, allowing the model to attend to more types of relationships simultaneously. The tradeoff is: more heads = more diverse attention patterns but each head has less representational capacity. Empirically, 8 heads with $d_k = 64$ and 16 heads with $d_k = 32$ often perform similarly, suggesting that the benefit of more heads roughly compensates for the reduced per-head capacity.
**Q3.** The CDR-head correspondence suggests that antibody binding has approximately 6 independent contact channels. If you were designing a transformer encoder specifically for antibody sequences, would you set $H = 6$?
*Answer:* Not necessarily. The correspondence is structural, not prescriptive. The number of heads $H$ is a hyperparameter that should be determined empirically. The CDR correspondence suggests that $H \geq 6$ is reasonable (to have at least as many attention channels as physical contact regions), but having more heads (e.g., $H = 8$ or $H = 12$) allows the model to capture additional relationship types beyond CDR-epitope contacts (e.g., framework-CDR interactions, intra-chain contacts). CALM uses a standard transformer encoder without constraining $H$ to equal 6. The correspondence provides intuition for why multi-head attention is well-suited to molecular recognition, not a design prescription.
**Q4.** In the worked example, Head 1 and Head 2 produce different attention patterns. What would happen if $W_Q^1 = W_Q^2$, $W_K^1 = W_K^2$, and $W_V^1 = W_V^2$ (identical projection matrices)?
*Answer:* Both heads would compute identical attention patterns and produce identical outputs. The concatenation would just be two copies of the same vector, and the multi-head attention would degenerate to single-head attention (with redundant parameters). During training, the gradient signal encourages the heads to diversify — if two heads are identical, they provide redundant information, and the gradient pushes their projections apart to capture complementary features. This is an emergent property of end-to-end training, not an explicit constraint.

## 10. Connection to Future Lectures

- **Lecture 3:** In CLIP and CALM, the dual encoders use multi-head self-attention within each molecular sequence. The encoder learns which residue-residue relationships are most informative for binding through the attention patterns in each head.
- **Lecture 6:** The convergence equation governs each attention head independently — each head implements a Boltzmann selection over its subspace. Multi-head attention computes $H$ parallel Boltzmann selections, each over a different projection of the molecular features.
- **Lecture 7:** The SFM architecture prescribes multi-head attention as the mechanism for capturing partially independent contact regions. Multi-head cross-attention in the decoder corresponds to position-specific contact probabilities across multiple binding modes.
- **Lecture 8:** The Level 1/2/3 classification applies to each attention head: each head implements softmax (the convergence equation) over its projected subspace. The correspondence between attention heads and CDR loops is discussed as evidence for Level 3 alignment at the sub-architecture level.